

CHAPTER II.4-5, WEEK 6

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2.16. Let X be a scheme, let $f \in \Gamma(X, \mathcal{O}_X)$, and define

$$X_f := \{x \in X : f_x \notin \mathfrak{m}_x \subset \mathcal{O}_{X,x}\}.$$

- (a) If $U = \text{Spec } B$ is an open affine subscheme of X , and if $\bar{f} \in B = \Gamma(U, \mathcal{O}_X|_U)$ is the restriction of f , show that $U \cap X_f = D(\bar{f})$. Conclude that X_f is an open subset of X .
- (b) Assume that X is quasi-compact. Let $A = \Gamma(X, \mathcal{O}_X)$, and let $a \in A$ be an element whose restriction to X_f is 0. Show that, for some $n > 0$, $f^n a = 0$.
- (c) Now assume that X has a finite cover by open affines U_i such that each intersection $U_i \cap U_j$ is quasi-compact. (This hypothesis is satisfied, for example, if $\text{sp}(X)$ is noetherian.) Let $b \in \Gamma(X_f, \mathcal{O}_{X_f})$. Show that, for some $n > 0$, $f^n b$ is the restriction of an element of A .
- (d) With the hypothesis of (c), conclude that $\Gamma(X_f, \mathcal{O}_{X_f}) \cong A_f$.

Proof. (a) $\mathfrak{p} \in U \cap X_f$ iff $\bar{f}_{\mathfrak{p}} \notin \mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}\mathcal{O}_{U,\mathfrak{p}}$, i.e. $\bar{f} \notin \mathfrak{p}$. It follows that $U \cap X_f = D(\bar{f})$ is open, so that X_f is a union of open sets, hence open itself.

(b) Let $a \in A$ with $a|_{X_f} = 0$. Then, for any open affine U intersection X_f , we define denote the restriction $\bar{a} := a|_U$. Then $\bar{a}|_{D(\bar{f})} = a|_{U \cap X_f} = 0$, whence there exists $n > 0$ such that $\bar{f}^n \bar{a} = \bar{f}^n a = 0$. Quasi-compactness of X gives us a finite affine open cover $\{U_i\}$, so that we can take some fixed $n > 0$ with $(f^n a)|_{U_i} = 0$ for all U_i . The sheaf axiom gives us that in fact $f^n a = 0$, so we are done.

(c) Consider $f_i := f|_{U_i}$ and $b_i := b|_{D(f_i)}$. Then, for each i , (a) gives us some $m > 0$ such that

$$(f_i^m b_i)|_{D(f_i)} = c_i|_{D(f_i)},$$

with $c_i \in \Gamma(U_i, \mathcal{O}_{U_i})$. Denote $U_{ij} = U_i \cap U_j$. $(c_i - c_j)|_{U_{ij} \cap X_f} = 0$, so (c) gives us $l > 0$ such that $f^l(c_i - c_j)|_{U_{ij}} = 0$, i.e. $(f^l c_i)|_{U_{ij}} = (f^l c_j)|_{U_{ij}}$. There are only finitely many U_{ij} , so we can take l such that the above holds for each i, j . Then we have $n = m + l > 0$ such that

$$\begin{aligned} (f_i^n b_i)|_{D(f_i)} &= c_i|_{D(f_i)}, \\ c_i|_{U_{ij}} &= c_j|_{U_{ij}} \end{aligned}$$

for all i, j . Then the c_i glue to an element $c \in A$ such that $c|_{X_f}$ is the gluing of the $f_i^n b_i|_{D(f_i)}$; namely, $f^n b$.

(d) By (c), we have a map $\phi : \Gamma(X_f, \mathcal{O}_{X_f}) \rightarrow A_f : b \mapsto f^n b / f^n$, which is well defined regardless of our choice of $n > 0$.¹ Suppose that $\phi(b) = 0$, so that there exists $m > 0$ with $f^{n+m} b = f^m(f^n b \cdot 1 - f^n \cdot 0) = 0$. Then, with notation as in (c), $f_i^{n+m} b_i = 0$ means $b_i = 0$ for each i , whence $b = 0$, showing injectivity.

¹We are often somewhat sloppy in writing $f^n b$ for the element of A which restricts to it, but we attempt to distinguish when necessary.

Finally, we show surjectivity. Fix $a/f^m \in A_f$. It suffices to find $b \in \Gamma(X_f, \mathcal{O}_{X_f})$ such that $f^{n+m}b - f^na = 0$ for $f^nb \in A$.² Fortunately, we need only glue together the maps $b_i = a|_{D(f_i)}/f_i^m$ to a map $b \in \Gamma(X_f, \mathcal{O}_{X_f})$, where compatibility is simple. Then $f^mb = a$, so that $\phi(b) = f^mb/f^m = a/f^n$, as desired. We conclude that ϕ is an isomorphism of rings. $\square$

2.17. A Criterion for Affineness

- (a) Let $f : X \rightarrow Y$ be a morphism of schemes, and suppose that Y can be covered by open subsets U_i such that for each i , the induced map $f^{-1}(U_i) \rightarrow U_i$ is an isomorphism. Then f is an isomorphism.
- (b) A scheme X is affine if and only if there is a finite set of elements $f_1, \dots, f_r \in A = \Gamma(X, \mathcal{O}_X)$ such that the open subsets X_{f_i} are affine, and $f_1, \dots, f_r$ generate the unit ideal in A .

Proof. (a) We have that

(b) If X is affine, then we can take $(f_1, \dots, f_r) = (1)$ to be a finite set of generators of the unit ideal, so that $X_{f_i} = D(f_i)$ are distinguished affines covering X .

Conversely, take a scheme X with such elements $f_1, \dots, f_r \in A$. Each X_{f_i} is affine and thus quasi-compact, so a finite cover of X by quasi-compact open sets implies X is quasi-compact. Each intersection $X_{f_i} \cap X_{f_j} = D_{X_{f_i}}(f_j)$ is affine and thus quasi-compact, so that 2.16(d) gives us $\Gamma(X_{f_i}, \mathcal{O}_{X_{f_i}}) \cong A_{f_i}$.

Exc. 2.4 supplies us with an isomorphism $\phi_i : X_{f_i} \xrightarrow{\sim} \text{Spec } A_{f_i}$, so that we have a cover of $\text{Spec } A$ by $D(f_i) \cong \text{Spec } A_{f_i}$ where ϕ $\square$

4.9. Show that a composition of projective morphisms is projective. Conclude that projective morphisms have all properties listed in Ex. 4.8.

Proof. We begin by showing that the Segre embedding $f : \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m \rightarrow \mathbb{P}_{\mathbb{Z}}^N$, with $N = (n+1)(m+1) - 1$, is a closed embedding.³ Recall that $\mathbb{P}_{\mathbb{Z}}^n = \text{Proj}(\mathbb{Z}[x_1, \dots, x_n])$, has an affine cover⁴

$$U_i = D(x_i) \cong \text{Spec}(\mathbb{P}_{\mathbb{Z}}^n)_{(x_i)} \cong \text{Spec } \mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}],$$

as we expect from the study of varieties. Then we have an open affine cover of $\mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m$ by

$$\begin{aligned} U_i \times_{\mathbb{Z}} V_j &\cong \text{Spec}(\mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}] \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{y_0}{y_j}, \dots, \frac{y_m}{y_j}]) \\ &\cong \text{Spec } \mathbb{Z}[\frac{x_k y_l}{x_i y_j} : 0 \leq k \leq n, 0 \leq l \leq m]. \end{aligned}$$

Then choose some bijection $\sigma : \{0, \dots, N\} \rightarrow \{(k, l)\}$, so we have surjective ring maps

$$\begin{aligned} \varphi_p : \mathbb{Z}[\frac{x_0}{x_p}, \dots, \frac{x_N}{x_p}] &\longrightarrow \mathbb{Z}[\frac{x_k y_l}{x_i y_j} : k, l] \\ \frac{x_q}{x_p} &\longmapsto \frac{x_{\sigma_1(q)} y_{\sigma_2(q)}}{x_{\sigma_1(p)} y_{\sigma_2(p)}} \end{aligned}$$

for each $0 \leq p \leq N$.

Surjectivity gives us corresponding closed immersions $f_p : U_{\sigma_1(p)} \cap V_{\sigma_2(p)} \rightarrow W_p$ by Ex. 2.18. It is not difficult to show these maps are compatible by considering the ring morphisms, so

²It is in fact equivalent, as if $f^{n+m+l}b - f^{n+l}a = 0$ for $l > 0$, then we can replace $n > 0$ by $n + l > 0$.

³Please tell me there is a much more elegant way of doing this.

⁴We do not omit the trivial term x_i/x_i , as it simply adjoins 1.

they glue together to our desired map $f : \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m \rightarrow \mathbb{P}_{\mathbb{Z}}^N$. It suffices to show that being a closed immersion is an affine local condition.

There is a cover of $\mathbb{P}_{\mathbb{Z}}^N$ by affine open sets upon which $f^\#$ is surjective, so $f^\#$ must be surjective. It suffices to show that it is a homeomorphism onto its image. Its image is the union of the finitely many closed sets $\text{im } f_p \subset \mathbb{P}_{\mathbb{Z}}^N$, and is thus closed. f is locally a homeomorphism onto its image, and it is probably injective.⁵ Thus, f is a closed embedding of schemes. As a result, any base extension f' of f is a closed embedding.

Now, if $g : X \rightarrow Y$ and $h : Y \rightarrow Z$ are projective maps, then we have factorizations

$$g : X \xrightarrow{\tilde{g}} Y \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m \rightarrow Y$$

$$h : Y \xrightarrow{\tilde{h}} Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n \rightarrow Z$$

for some $m, n > 0$, where $\tilde{g}, \tilde{h}$ are closed immersions. The pullback of a closed immersion is a closed immersion, so we consider the following pullback square.

$$\begin{array}{ccc} Y \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m & \xrightarrow{\tilde{h}'} & Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\tilde{h}} & Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n \end{array}$$

Combining this with our closed immersion $f' : Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m \rightarrow Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^N$ from earlier, we get the following diagram.

$$\begin{array}{ccccccc} X & \longrightarrow & Y \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n & \xrightarrow{\tilde{h}'} & Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m & \xrightarrow{f'} & Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^N \\ & \searrow g & \downarrow & & \downarrow & & \swarrow \\ & & Y & \xrightarrow{\tilde{h}} & Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n & & \\ & & & \searrow h & \downarrow & & \\ & & & & Z & & \end{array}$$

The top line is a composition of closed immersions, and thus is a closed immersion which composes with the projection onto Z to give $h \circ g$. Thus, we have a factorization $h \circ g : X \rightarrow Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^N \rightarrow Z$, showing $h \circ g$ is projective.

By Exc. 4.8, it suffices to show that closed immersions and base extensions of projective maps are projective. First, take $g : X \rightarrow Y$ be a closed immersion. We have the following diagram.

$$\begin{array}{ccccc} X & \longrightarrow & X \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n & \xrightarrow{g'} & Y \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n \\ & \searrow & \downarrow & & \downarrow \\ & & X & \xrightarrow{g} & Y \end{array}$$

g' is a closed immersion, so it suffices to show that $X \rightarrow X \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n$ is a closed immersion. Being a closed immersion is an affine local condition, so it suffices to show that $\text{Spec } A \rightarrow \text{Spec } A \times_{\mathbb{Z}} U_i$ is a closed immersion. $U_i \cong \text{Spec } \mathbb{Z}[x_0/x_i, \dots, x_n/x_i]$, so this amounts to showing

$$A[x_1, \dots, x_n] \cong A \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}\right] \longrightarrow A$$

⁵How do you actually show injectivity? I feel like it should be obvious.

is surjective, which is clear.

Now, suppose $g : X \rightarrow Y$ is projective, and $Z \rightarrow Y$ is a base extension. Then we have the following diagram.

$$\begin{array}{ccccc}
 & & g' & & \\
 & \nearrow & & \searrow & \\
 X' & \longrightarrow & Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n & \longrightarrow & Z \\
 \downarrow & & \downarrow & & \downarrow \\
 X & \longrightarrow & Y \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n & \longrightarrow & Y \\
 & \searrow & & \nearrow & \\
 & & g & &
 \end{array}$$

Then $X' \rightarrow Z \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n$ is a closed immersion, so that $g' : X' \rightarrow Z$ factors as desired. We conclude that base extensions of projective maps are projective, whence Exc. 2.8 gives us the other three properties. Namely, a product of projective morphisms is projective, $g \circ f$ projective with g separated implies f projective, and the reduction f_{red} of a projective map f is projective. $\square$

5.1. Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{E}$ be a locally free $\mathcal{O}_X$ -module of finite rank. We define the *dual* of $\mathcal{E}$ to be the sheaf $\mathcal{E}^\vee := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$.

- (a) Show that $(\mathcal{E}^\vee)^\vee \cong \mathcal{E}$.
- (b) For any $\mathcal{O}_X$ -module $\mathcal{F}$, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F}) \cong \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F}$.
- (c) For any $\mathcal{O}_X$ -modules $\mathcal{F}, \mathcal{G}$,

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G})).$$

- (d) (*Projection Formula*). If $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces, $\mathcal{F}$ is an $\mathcal{O}_X$ -module, and $\mathcal{E}$ is a locally free $\mathcal{O}_Y$ -module of finite rank, then there is a natural isomorphism

$$f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^* \mathcal{E}) \cong f_*(\mathcal{F}) \otimes_{\mathcal{O}_Y} \mathcal{E}.$$

Proof. $\square$

5.3. Let $X = \text{Spec } A$ be an affine scheme. Show that the functors $\widetilde{}$ and Γ are adjoint. That is, for any A -module M , and for any sheaf of $\mathcal{O}_X$ -modules $\mathcal{F}$, there is a natural isomorphism

$$\text{Hom}_A(M, \Gamma(X, \mathcal{F})) \cong \text{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathcal{F})$$

Proof. $\square$

5.5. Let $f : X \rightarrow Y$ be a morphism of schemes.

- (a) Show by example that if $\mathcal{F}$ is coherent on X , then $f_* \mathcal{F}$ need not be coherent on Y , even if X and Y are varieties over a field k .
- (b) Show that a closed immersion is a finite morphism.
- (c) If f is a finite morphism of noetherian schemes, and if $\mathcal{F}$ is coherent on X , then $f_* \mathcal{F}$ is coherent on Y .

5.6. *Support.* Recall the notions of support of a section of sheaf, support of a sheaf, and subsheaf with supports from (Ex. 1.14) and (Ex. 1.20).

- (a) Let A be a ring. Let M be an A -module, let $X = \text{Spec } A$, and let $\mathcal{F} = \widetilde{M}$. For any $m \in M = \Gamma(X, \mathcal{F})$, show that $\text{Supp } m = V(\text{Ann } m)$, where $\text{Ann } m = \{a \in A \mid am = 0\}$ is the *annihilator* of m .

- (b) Now suppose that A is noetherian, and M is finitely generated. Show that $\text{Supp } \mathcal{F} = V(\text{Ann } M)$.
- (c) The support of a coherent sheaf on a noetherian scheme is closed.
- (d) For any ideal $\mathfrak{a} \subset A$, we define a submodule $\Gamma_{\mathfrak{a}}(M)$ of M by

$$\Gamma_{\mathfrak{a}}(M) = \{m \in M \mid \exists n > 0 (\mathfrak{a}^n m = 0)\}.$$

Assume that A is noetherian, and M is any A -module. Show that $\widetilde{\Gamma_{\mathfrak{a}}(M)} \cong \mathcal{H}_Z^0(\mathcal{F})$, where $Z = V(\mathfrak{a})$ and $\mathcal{F} = \widetilde{M}$.

- (e) Let X be a noetherian scheme, and let Z be a closed subset. If $\mathcal{F}$ is a quasi-coherent (resp., coherent) $\mathcal{O}_X$ -module, then $\mathcal{H}_Z^0(\mathcal{F})$ is also quasi-coherent (resp., coherent).

5.8. Again let X be a noetherian scheme, and $\mathcal{F}$ a coherent sheaf on X . We will consider the function

$$\varphi(x) = \dim_{k(x)} \mathcal{F}_x \otimes_{\mathcal{O}_X} k(x),$$

where $k(x) = \mathcal{O}_x / m_x$ is the residue field at the point $x \in X$. Use Nakayama's lemma to prove the following results.

- (a) The function φ is *upper semi-continuous*, i.e., for any $n \in \mathbb{Z}$, the set $\{x \in X \mid \varphi(x) \geq n\}$ is closed.
- (b) If $\mathcal{F}$ is locally free and X is connected, then φ is a constant function.
- (c) Conversely, if X is reduced and φ is constant, then $\mathcal{F}$ is locally free.

5.16. For $(X, \mathcal{O}_X)$ a ringed space and $\mathcal{F}$ a sheaf of $\mathcal{O}_X$ -modules, we define the *tensor algebra*, *symmetric algebra*, and *exterior algebra* of $\mathcal{F}$ by taking the sheaves associated to the associated presheaf.

- (a) Suppose that $\mathcal{F}$ is locally free of rank n . Then $T^r(\mathcal{F})$, $S^r(\mathcal{F})$ and $\wedge^r(\mathcal{F})$ are locally free, of ranks rn , $\binom{n+r-1}{n-1}$, and $\binom{n}{r}$ respectively.
- (b) Again, let $\mathcal{F}$ be locally free of rank n . Then the multiplication map

5.17. *Affine morphisms.* A morphism $f : X \rightarrow Y$ is *affine* if there is an affine open cover $\{V_i\}$ of Y such that $f^{-1}(V_i)$ is affine for each i .

- (a) Show that f is an affine morphism if and only if for each every open affine $V \subset Y$, $f^{-1}(V)$ is affine.
- (b) An affine morphism is quasi compact and separated. Any finite morphism is affine.
- (c) Let Y be a scheme, and let $\mathcal{A}$ be a quasi-coherent sheaf of $\mathcal{O}_Y$ -algebras. Show that there is a unique scheme X , and a morphism $f : X \rightarrow Y$, such that for every open affine $V \subset Y$, $f^{-1}(V) \cong \text{Spec } \mathcal{A}(V)$, and for every inclusion $U \hookrightarrow V$ of open affines of Y , the morphism $f^{-1}(U) \hookrightarrow f^{-1}(V)$ corresponds to the restriction homomorphism $\mathcal{A}(V) \rightarrow \mathcal{A}(U)$. The scheme X is called **Spec** $\mathcal{A}$.
- (d) If $\mathcal{A}$ is a quasi-coherent $\mathcal{O}_Y$ -algebra, then $f : X = \text{Spec } \mathcal{A} \rightarrow Y$ is an affine morphism and $\mathcal{A} \cong f_* \mathcal{O}_X$. Conversely, if $f : X \rightarrow Y$ is an affine morphism, then $\mathcal{A} = f_* \mathcal{O}_X$ is a quasi-coherent sheaf of $\mathcal{O}_Y$ -modules, and $X \cong \text{Spec } \mathcal{A}$.
- (e) Let $f : X \rightarrow Y$ be an affine morphism, and let $\mathcal{A} = f_* \mathcal{O}_X$. Show that f_* induces an equivalence of categories from the category of quasi-coherent $\mathcal{O}_X$ -modules to the category of quasi-coherent $\mathcal{A}$ -modules (i.e., quasi-coherent $\mathcal{O}_Y$ -modules having the structure of an $\mathcal{A}$ -module.) [*Hint*: For any quasi-coherent $\mathcal{A}$ -module

$\mathcal{M}$, construct a quasi-coherent $\mathcal{O}_X$ -module $\widetilde{\mathcal{M}}$, and show that the functors f_* and $\widetilde{\cdot}$ are inverse to each other.]

Proof. (a) It suffices to show that an affine morphism has this property. Take $f : X \rightarrow Y$ affine with open affine cover $\{V_i\}$ such that each $f^{-1}(V_i)$ is affine, and let $V \subset Y$ be open affine. Affine communication lemma gives us a cover of V by $D(g)$, $g \in \mathcal{O}_Y(V_i)$ which are distinguished open in both V and V_i for some i . Note that each $f^{-1}(D(g_\alpha)) = D(f^\#(V_i)(g))$ is open affine. Then we may assume $V = Y$ is affine, and that each V_i is a distinguished open set of Y .

We have a cover of X by open affines $U_i = f^{-1}(V_i)$

(b) Let $f : X \rightarrow Y$ be affine. Then, for any $V \subset Y$ affine open, $f^{-1}(V)$ is affine by (a), and thus quasi-compact. It follows from 3.2 that f is quasi compact.

Taking an affine cover V_i of Y , we have an affine cover $U_i := f^{-1}(V_i)$ of X . Then $f|_{U_i} : U_i \rightarrow V_i$ is a morphism of affine schemes, hence separated, so that $\Delta|_{U_i} : U_i \rightarrow U_i \times_{V_i} U_i$ is a closed immersion. It follows that $\Delta : X \rightarrow X \times_Y X$ is in fact a closed immersion, i.e. f is separated.

If f is finite, then there is an affine cover $V_i \cong \text{Spec } B_i$ of Y such that $f^{-1}(V_i) \cong \text{Spec } A_i$ such that A_i is a finitely generated B_i module. In particular, $f^{-1}(V_i)$ is affine.

(c)

□

5.18. *Vector Bundles.* Let Y be a scheme. A *(geometric) vector bundle* of rank n over Y is a scheme X and a morphism $f : X \rightarrow Y$, together with additional data consisting of an open covering $\{U_i\}$ of Y , and isomorphisms $\psi_i : f^{-1}(U_i) \rightarrow \mathbb{A}_{U_i}^n$, such that for any i, j , and for any open affine subset $V = \text{Spec } A \subset U_i \cap U_j$, the automorphism $\psi = \psi_j \circ \psi_i^{-1}$ of $\mathbb{A}_V^n = \text{Spec } A[x_1, \dots, x_n]$ is given by a